Housing Price Analysis Report

Introduction

In this project, I conducted an Exploratory Data Analysis (EDA) on a housing dataset containing 545 residential properties and 13 variables that describe various housing characteristics and their selling prices. The aim of this analysis was to understand the factors that influence house prices and identify trends that could be useful for both buyers and sellers. The dataset includes variables such as price, area, number of bedrooms and bathrooms, stories, furnishing status, air conditioning, and location-related features.

Before beginning the analysis, I assessed the dataset for data quality issues. I found that there were no missing values and no duplicate records, indicating that the dataset was already clean and well-structured. I also checked for outliers in the price and area columns and found a few extreme values. However, these appeared to represent genuinely large or expensive properties rather than data-entry errors, so I retained them for the analysis.

Key Findings

One of the first relationships I investigated was between house size and price. I found a correlation coefficient of approximately 0.536 between area and price, indicating a moderate positive relationship. This suggests that larger houses generally sell for higher prices, although house size alone does not fully explain property values.

Location was found to be one of the strongest factors affecting price, houses located on the main road had an average price of approximately 4.99 million, compared to about 3.40 million for houses located away from the main road. Similarly, properties situated in preferred areas had an average price of approximately 5.88 million, compared to 4.43 million for properties outside these areas. This means that houses in preferred areas command a premium of roughly 1.45 million on average.

I also discovered that certain property features significantly increase house prices. For example, houses with air conditioning sold for an average price of approximately 6.01 million, while those without air conditioning averaged only 4.19 million. Additionally, houses that had both a basement and air conditioning recorded an average price of about 6.08 million, further demonstrating the value buyers place on comfort and additional amenities.

Another notable finding was the impact of furnishing status. Furnished houses recorded the highest average price at approximately 5.50 million, followed by semi-furnished houses at 4.91 million. Unfurnished houses had the lowest average price at approximately 4.01 million. This suggests that buyers are willing to pay more for homes that require little or no additional setup.

Finally, I observed a clear upward trend between the number of stories and house prices. Average prices increased steadily from approximately 4.17 million for single-story houses to over 7.20 million for four-story houses. This indicates that larger multi-level properties generally command higher market values.

Data Visualisations

The histogram (figure 1) shows that house prices are right-skewed, with most properties concentrated in the lower to middle price ranges and a small number of high-value properties creating a long upper tail. A small number of luxury properties created a long right tail in the distribution. This observation helps explain why the mean house price is higher than the median price, indicating the influence of high-value properties on the overall average.

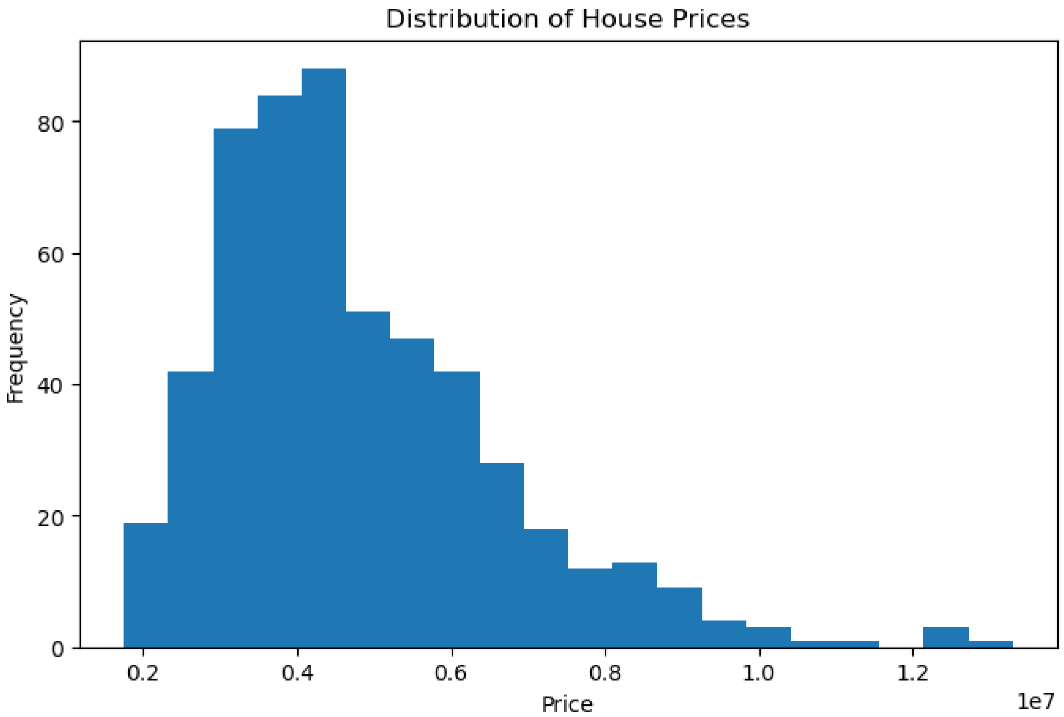

Figure 1: Distribution of House Prices (Histogram)

The scatter plot (Figure 2) demonstrates a positive relationship between area and price. As house area increased, prices generally increased as well. However, I noticed considerable variation among houses with similar sizes, suggesting that other factors such as location, furnishing status, and amenities also contribute significantly to house prices.

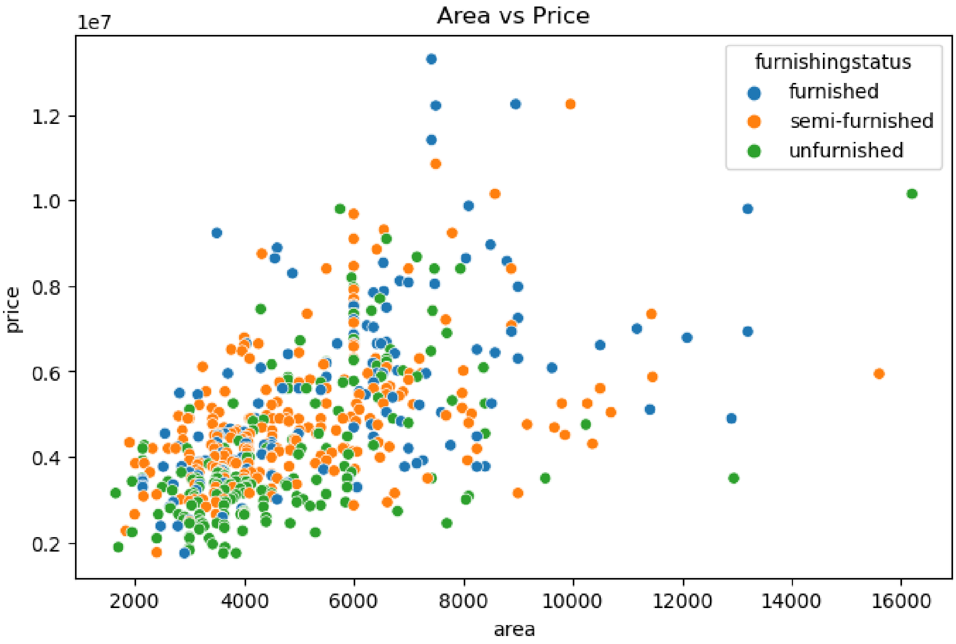

Figure 2: Area vs Price (Scatter Plot)

The box plot (figure 3) shows that houses with air conditioning have a noticeably higher median price than those without it. In addition, the upper quartile and maximum prices for air-conditioned homes are substantially higher, suggesting that air conditioning is often associated with premium properties. The wider spread among air-conditioned houses also indicates greater variation in prices within this group.

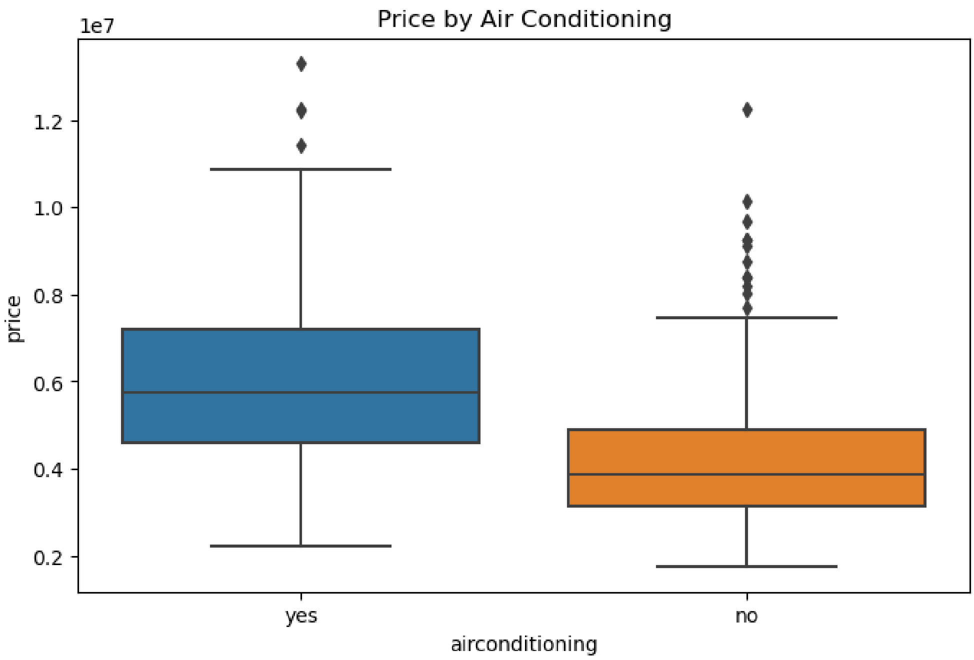

Figure 3: Price by Air Conditioning

The bar chart (figure 4) demonstrates that average house prices generally increase as the number of bedrooms increases. Houses with more bedrooms provide greater living space and are therefore valued more highly by buyers. However, I observed a slight decline in average price for 6-bedroom houses. This may be due to the relatively small number of 6-bedroom properties in the dataset or differences in other influential features such as location, furnishing status, and property condition. Therefore, while bedroom count contributes to house value, it should not be considered in isolation when evaluating property prices.

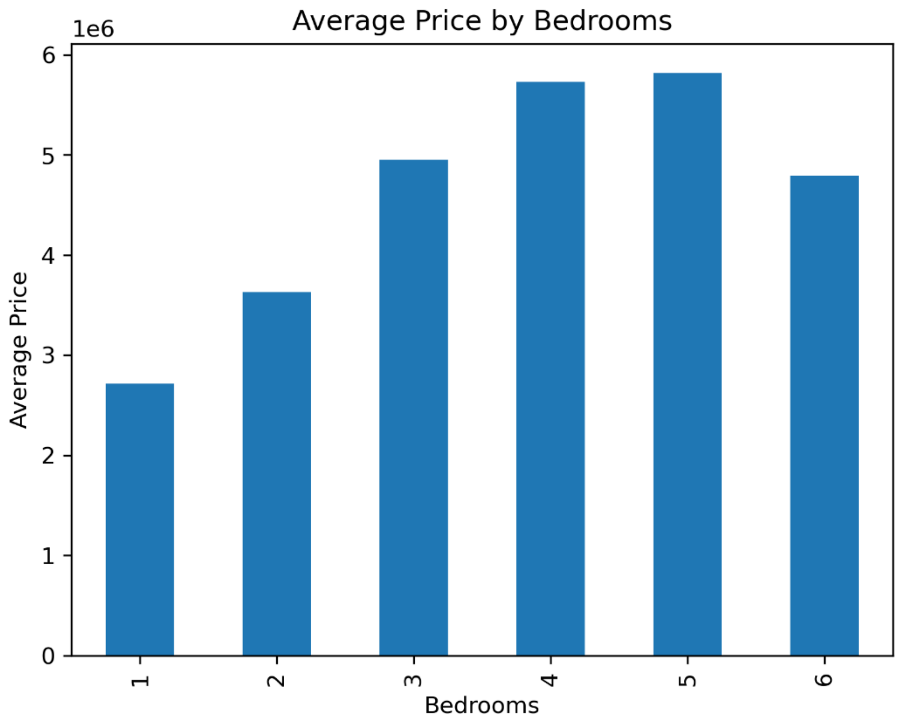

Figure 4: Average Price by Number of Bedrooms (Bar Chart)

The heatmap (figure 5) provided a comprehensive view of the relationships between numerical variables in the dataset. I observed that price had its strongest positive correlation with area, followed by variables such as bathrooms and stories. The heatmap also showed that no single factor perfectly explains house prices, highlighting the importance of considering multiple features when evaluating a property.

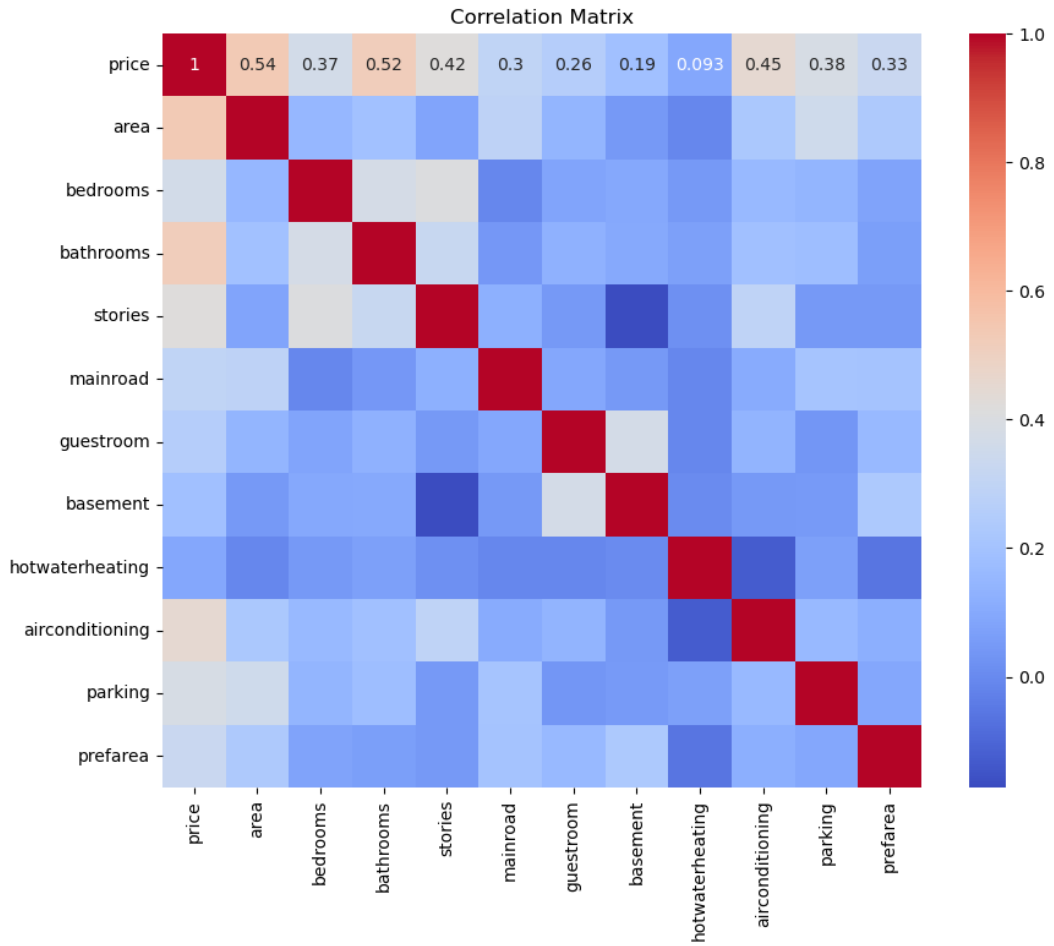

Figure 5: Correlation Matrix (Heatmap)

Conclusion and Recommendations

Based on my analysis, I conclude that house prices are influenced by a combination of size, location, and property features. Area, preferred location, air conditioning, furnishing status, and the number of stories all contribute positively to property value.

For buyers seeking good value for money, I would recommend considering properties outside preferred areas, as they tend to be significantly cheaper while still offering comparable living space. I would also suggest exploring semi-furnished houses, which provide a balance between affordability and convenience. Finally, buyers should carefully assess whether premium features such as air conditioning and additional stories justify the extra cost based on their personal needs and budget.